

Chapter 35 — Urologic Evaluation of the Child

TRIAGE OF COMMON PAEDIATRIC UROLOGY CONSULTATIONS

- **Spectrum:** simple problems (communicating hydrocele, nocturnal enuresis) to complex anomalies — posterior urethral valves (PUV), bladder exstrophy, differences of sexual differentiation (DSD).
- **Lifelong consequences:** many decisions affect the patient for life, including fertility.
- **Guardians share decision making;** instrumentation and surgical options are **constrained by age and size**.
- **Acuity set from child, parents and referrer** → emergency department vs routine clinic (Table 35.1).

TABLE 35.1 COMMON PEDIATRIC UROLOGIC CONDITIONS AND THEIR LEVELS OF ACUITY

Acuity	Conditions
Emergent	Acute abdomen; acute kidney injury (AKI); acute postoperative problem; acute scrotum; acute urinary retention; incarcerated inguinal hernia; newborn with ambiguous genitalia, bladder/cloacal exstrophy or renal vein thrombosis; obstructive uropathy with urinary tract infection (UTI) and/or AKI; paraphimosis; priapism; trauma
Urgent (within 24 h)	Gross haematuria without infection; newborn bilateral hydronephrosis or solitary kidney with hydronephrosis (can be emergent); kidney tumour; testicular tumour; spina bifida
Semiurgent (within 1–3 wk)	Afebrile, symptom-controlled urolithiasis; amenorrhoea in adolescent female; consultation after treated febrile UTI; interlabial mass; postnatal evaluation of urinary tract dilation (excluding suspected bladder outlet obstruction); reducible inguinal hernia
Routine	Asymptomatic hydrocele; circumcision evaluation; daytime urinary incontinence/nocturnal enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testicle; varicocele

- **What it is for:** tiers reflect **what delay costs** — an organ (torsion, infected obstruction), a life (salt-wasting crisis) or nothing time-critical.
- **Pattern: context shifts the tier** — hernia emergent if incarcerated, semiurgent if reducible; hydronephrosis urgent if bilateral/solitary, semiurgent otherwise.
- **Trap:** gross haematuria is urgent only “**in the absence of infection**”; microscopic haematuria is routine.

Emergent referrals

Acute scrotum

- **Definition:** new-onset pain, swelling and/or tenderness of the intrascrotal contents.
- **Commonest diagnoses:** testicular appendage torsion, spermatic cord torsion, epididymitis. **Most time-sensitive = spermatic cord torsion** (delay → testicular loss).

Testicular appendage torsion.

- **Commonest cause of acute scrotum;** mostly **prepubertal;** **supportive care.**
- Early, before diffuse inflammation → **point tenderness** over the torsed appendix.
- Ultrasonography **rarely shows the appendage itself** — shows **epididymal hyperaemia** (mimics epididymitis).

Spermatic cord torsion.

- **Prevalence ≈ 1 in 4000 males;** any age, **majority after age 10.**
- **Acute severe pain,** frequently with **nausea and vomiting.**
- **Commonest signs:** generalised testicular tenderness, abnormal testicular orientation, absent cremasteric reflex. **A present cremasteric reflex does not exclude torsion.**

- **Clinical diagnosis** — if suspected, **forgo imaging → immediate exploration; viability is inversely related to duration.**
- **Manual detorsion** may relieve symptoms without fully untwisting the cord → **does not obviate urgent exploration.**
- At detorsion/orchiopexy → **contralateral orchiopexy** to prevent metachronous torsion.

TWIST (TESTICULAR WORKUP FOR ISCHEMIA AND SUSPECTED TORSION) SCORE — MAXIMUM 7

Item	Points	Score band	Action
Testicular swelling	2	0 — low risk	No ultrasonography needed
Hard testicle	2	1-5 — intermediate	Scrotal ultrasonography
Absent cremasteric reflex	1	≥6 — high risk	Immediate exploration without ultrasonography
Nausea/vomiting	1		
High-riding testicle	1		

- **Pattern:** the 2-point items (swelling, hardness) are **signs of an ischaemic testis**; 6-7 needs both plus most 1-point items.

Neonatal (extravaginal) torsion.

- Can occur **before birth** → usually **testicular nubbin / vanishing testis.**
- Signs: **enlarged, firm testis without apparent pain** or distress.
- Acute perinatal diagnosis → **elective vs urgent exploration is controversial**; authors prefer **urgent** intervention.

Ambiguous genitalia in the newborn

- **Emergency** on three counts: **medical stabilisation, diagnosis, assignment of sex of rearing** (DSD in detail: Chapters 59-60).
- Stabilisation = recognise/prevent/treat **salt-wasting congenital adrenal hyperplasia (CAH)** — **commonest cause of newborn ambiguous genitalia**; commonest type **21-hydroxylase deficiency** (≈95%).
- **Three presentations:** virilisation + salt wasting; simple virilisation without salt wasting (ambiguous genitalia in females; sexual precocity at 2-3 years in males); neither.
- **Classic virilised 46,XX:** no palpable gonads, possible clitoromegaly, common urogenital sinus.
- **Salt wasting (either sex):** failure to regain birth weight, weight loss, dehydration → death from **hyperkalaemia, dehydration, shock.**
- **Crisis usually begins day 4-15** → **do not discharge early** if suspected.
- **Essential tests:** **17-hydroxyprogesterone + serum electrolytes.** Newborn screen checks **only 21-hydroxylase** and **may not return before discharge.**
- Treatment: **lifelong glucocorticoid + mineralocorticoid.** Other CAH variants (not screened) may lack salt wasting but still cause **adrenal crisis.**

Bladder/cloacal exstrophy

- **Rare** (≈2.15 per 100,000 US live births); spectrum **epispadias → bladder exstrophy → cloacal exstrophy** (cloacal rarer still; multidisciplinary).
- Bladder mucosa at birth: **pink, healthy but delicate.**
- Protection sequence: **tie cord with 2-0 silk** (plastic clamp damages mucosa) → **cover with nonadherent film** (cling film/Tegaderm) → at each nappy change **gentle saline irrigation + new wrap.**
- **Immediate transfer to a paediatric centre** with full diagnostic and specialist services.

- Initial examination: **bladder plate size/quality, pubic diastasis, urethral plate length/width, penile length, associated anomalies.**

Obstructive uropathy with urinary tract infection and/or acute kidney injury

- **Commonest acute presentation** of obstruction with infection = **obstructing urolithiasis.**
- Active infection + obstruction → **prompt drainage: ureteral stent or nephrostomy.**
- Children **compensate for shock longer; hypotension may appear only when shock is decompensated.**
- Weigh **morbidity and quality-of-life cost of a nephrostomy in a small child** when choosing decompression.
- **Ureteropelvic junction (UPJ) / ureterovesical junction (UVJ) obstruction** with infection, extrinsic compression or AKI → consider **cutaneous pyelostomy or ureterostomy.**
- **Large ureterocele** → ureteral *and* bladder outlet obstruction → urethral catheter or prompt ureterocele decompression.
- Bladder outlet obstruction (e.g. PUV) → may need **cutaneous vesicostomy.**

Renal vein thrombosis

- **Rare** (≈2.6 per 100,000 live births); consider with **palpable flank mass and/or gross haematuria in a newborn.**
- **Male and left-sided predominance; two-thirds present within first 3 days.**
- **Risk factors = hypercoagulability:** prematurity, umbilical venous catheter, congenital heart disease, respiratory distress syndrome, maternal gestational diabetes, infection.
- **Classic triad: haematuria + flank mass + thrombocytopenia.**
- **Diagnosis: Doppler ultrasound; no evidence-based treatment guidelines.**
- Most infants → complete, cortical or segmental **infarction and/or hypertension** → ↑ chronic kidney disease (CKD) and death **regardless of treatment.**
- Treatment reported: supportive care → anticoagulation → thrombolysis; **surgery rarely warranted.**

Paraphimosis

- **Foreskin trapped behind the glans** in an uncircumcised male — **emergent.**
- **Pitfall:** swollen uncircumcised penis **mistaken for circumcised** → **always ask circumcision status.** Rarely occurs after circumcision if a **cicatrix** has formed.
- Severe **oedema within hours** → harder reduction; unreduced → **necrosis, gangrene.**
- **Differential: hair tourniquet** — circumcised or not; same emergent sequelae; inspect closely.
- **Reduction:** manual compression of glans + distal traction on oedematous foreskin; analgesia by topical anaesthetic, penile block or **ketamine procedural sedation.**
- **Adjuncts to reduce oedema:** ice pack 5 min; Coban wrap 15–30 min; granulated sugar 1–2 h; multiple needle punctures.
- Manual failure → **dorsal slit or circumcision under anaesthesia.**

Priapism

- **Definition:** full or partial erection **≥4 hours unrelated to sexual stimulation;** urological emergency; paediatric prevalence unknown, uncommon.

COMPARISON — ISCHAEMIC VS NONISCHAEMIC VS NEONATAL PRIAPISM

	Ischaemic (IP)	Nonischaemic	Neonatal
Frequency	By far commonest in children; only truly emergent type	Less common	Rare; mostly idiopathic
Mechanism	Little/no cavernous flow	Trauma (penile, perineal, pelvic) → high inflow, sinusoidal pooling	Presumed subclinical birth-canal perineal trauma ; IP not reported
Clinical	Rigid, painful	Often painless	Not painful
Corporal gas	Hypoxic (pO₂ <40 mm Hg), hypercapnic, acidotic, glucopenic	—	—
Management	Find cause + achieve detumescence (Chapter 55)	Initially expectant ; then compression, pharmacological, embolisation or ligation	Monitor ; safe up to 5 days (average 4–5 days); full recovery
Risk	Erectile dysfunction, fibrotic disfigurement	Natural history unclear	Benign

Ischaemic priapism.

- **Commonest cause: sickle cell disease** → other haemoglobinopathies, sickle cell trait, leukaemia, pelvic tumours, parenteral nutrition, Henoch–Schönlein purpura; **severe constipation** also reported.
- **Drugs:** phosphodiesterase-5 (PDE-5) inhibitors, antipsychotics, antidepressants, stimulants, antihypertensives, anaesthesia, recreational drugs (cocaine).
- Work-up: history, examination → consider **urinalysis + urine and plasma toxicology**.
- **Sickle cell precipitants:** nocturnal erections, sexual activity, dehydration, fever, cold exposure.
- Opioid analgesia sometimes needed (inhibits erection in animal models).
- **“First aid” while awaiting urology:** physical exercise (running up stairs), urination, cold bath, ejaculation, ↑ fluids.
- **Cold packs** — reduce pain, ischaemic damage, vasoconstrict — **only if sickle cell disease not suspected**; they can **induce** priapism in sickle cell disease.
- In hospital: **oxygen + intravenous fluids**, especially in haemoglobinopathy.
- **Refractory:** corporal blood gas confirms low-flow IP; **colour Doppler** is highly sensitive and an alternative.

Nonischaemic priapism. Children with perineal/penile trauma need a **primary and secondary survey**.

Trauma*Renal trauma.*

- **Kidney = commonest injured genitourinary organ** (detail: Chapter 62). Paediatric kidney **more susceptible**; congenital dilation (e.g. UPJ obstruction) commoner than in adults.
- **Blunt trauma by far commonest:** motor vehicle collision, falls, sports, pedestrian vs vehicle, increasingly **all-terrain vehicles**.
- Flank tenderness, ecchymosis, rib fractures as in adults — but **hypotension, tachycardia and haematuria are less reliable** → **mechanism of injury becomes the key predictor**.
- **2020 American Urological Association (AUA) urotrauma guideline** — image stable blunt trauma if **gross or microscopic haematuria + systolic BP <90 mm Hg, or concerning mechanism/examination** (rapid deceleration, significant flank blow, rib fracture, significant flank ecchymosis, penetrating flank/abdominal/lower chest injury).

- **CT preferred**; renal ultrasonography acceptable.
- **American Association for the Surgery of Trauma (AAST) renal Organ Injury Scale (OIS)** predicts intervention and complications in children as in adults → use identically.
- **Nonoperative management ideal**; intervention usually **minimally invasive** (stent, nephrostomy/drain, angioembolisation).

Straddle injury.

- Blow to perineum from striking an object with own body weight; **summer**; bicycle crossbars, ladders, pools, playgrounds.
- **Always consider sexual abuse** (≈1 in 7 males, 1 in 3 females abused in childhood).
- **Investigate for sexual assault if**: age <9 months; perianal, hymenal or vaginal injury; extensive/severe injury; concurrent non-urogenital injuries; **history inconsistent with findings**.

COMPARISON — STRADDLE INJURY: FEMALE VS MALE

	Female	Male
Epidemiology/ presentation	Commonest cause of female genital trauma (≈80%) ; perineal bleeding commonest symptom; vaginal/vulval laceration or haematoma	Scrotal or penile lacerations, abrasions, ecchymosis
Tissue factor	Unoestrogenised prepubertal tissue fragile, non-distensible → minor trauma bleeds impressively	—
Default	Without pelvic fracture → mostly expectant	Expectant or repair under sedation
Escalate	Suspected injury beyond labia, ongoing bleeding or inadequate examination → examination under anaesthesia (some manage with ED procedural sedation)	Significant scrotal haematoma → ultrasonography to exclude testicular rupture (immediate repair)
Urethral injury	Rare without pelvic fracture; ≈5% of female pelvic fractures . Suspect with gross haematuria, retention, vaginal/rectal injury → EUA + cystoscopy (imaging adds little). Acute: suprapubic tube + delayed repair ; or primary realignment over catheter within days if stable. Continence good if bladder neck spared	Suspect with gross haematuria, retention, meatal blood, perineal haematoma → do not attempt catheter → retrograde urethrogram (RUG), may need sedation/GA; cannot drain → suprapubic tube

- **Female long-term sequelae to prevent**: stricture, fistula, incontinence, sexual dysfunction (vaginal, hymenal, urethral or anal injury).

Blunt penile injury.

- Rare; in the USA often **toilet-seat crush** — mean age ≈3.9 years, almost all **at home**; long-term effects unknown.
- Heterogeneous; often **expectant**. **Gross haematuria, meatal blood or inability to void** → ultrasound, RUG or flexible cystoscopy to find surgical cases (penile fracture, urethral injury).

Acute urinary retention

- ED incidence ≈8.5–30 per 100,000 visits. **Bladder scanner or ultrasonography** estimates volume.
- **Expected bladder capacity (mL) = (age + 2) × 30** — distinguishes retention from reduced urine output.
- **Causes**: drugs; infection (UTI, balanoposthitis); post-instrumentation/surgery; neurogenic; trauma; functional (pelvic floor dysfunction, **constipation/faecal impaction**); benign or malignant mass.
- Work-up: examination, renal ultrasonography, abdominal radiograph → no cause found → **neurological evaluation ± spinal imaging**.
- Not all need a catheter: **oral analgesia, warm sitz bath, rectal disimpaction/enema** may induce voiding.

Urgent referrals to be evaluated within 24 hours

Newborn hydronephrosis (bilateral or solitary kidney)

- Mostly **prenatally detected**; some present with failure to thrive, feeding intolerance, abdominal mass or febrile UTI.
- **Postnatal ultrasound before discharge** if prenatal hydronephrosis (**third-trimester anteroposterior renal pelvic diameter >7 mm or Urinary Tract Dilation (UTD) A2–3**) in a **solitary kidney, bilateral, or with suspected bladder outlet obstruction**.
- **Timing: ideally after 48 h** (earlier scans **underestimate** dilation) — **but immediately if outlet obstruction suspected**.
- **Serum creatinine reflects maternal creatinine for 48 h** → follow for a **downward trend**.
- **Moderate–severe bilateral (UTD P2–P3)**: review prenatal images/history — **oligo-/anhydramnios and gestation of onset**, other anomalies; respiratory status, vitals, whether voided (**voiding ≠ normal emptying**).
- **Differential**: outlet obstruction (**PUV, urethral atresia, megalourethra, neurogenic bladder, obstructive ureterocele, hydrocolpos**); bilateral UPJ obstruction; vesicoureteral reflux (VUR) or UVJ obstruction.
- Examination: spinal, anorectal, genital anomalies, patent urachus; palpable kidney/bladder.
- **Male + bilateral hydronephrosis, thick bladder wall, hydroureter or dilated posterior urethra** → **voiding cystourethrogram (VCUG) before discharge**.
- High suspicion of outlet obstruction → **feeding tube/catheter per urethra** pending imaging, especially if unstable.
- Catheter impossible (urethral atresia) or supra-urethral obstruction → other decompression. **High-grade obstruction bilaterally/solitary kidney** → **vesicostomy, ureterostomy, nephrostomy or vaginostomy**; some suit **early reconstruction** (e.g. neonatal pyeloplasty for high-grade UPJ obstruction in a solitary kidney).
- Identify obstruction threatening **upper tract, infection or function**; **nephrology** if parenchymal abnormality or renal insufficiency.
- **Most need no neonatal intervention** but moderate–severe bilateral/solitary cases need **close follow-up**.
- **Educate every family**: neonatal fever → **properly collected urine**; also urine output change, haematuria, vomiting, poor feeding, palpable kidney/bladder, poor weight gain.

Gross haematuria

- **Definition**: visible blood with **red blood cells (RBCs) confirmed** on urinalysis/microscopy.
- **Discriminating symptoms**: flank/abdominal pain, fever, dysuria, incomplete emptying, trauma, joint pain, oedema, recent illness (viral cystitis, post-streptococcal glomerulonephritis); **character** — dark vs bright red, clots.
- **Baseline for all**: urinalysis + microscopy, urine culture, renal–bladder ultrasound.

COMPARISON — GLOMERULAR VS UROLOGICAL (LOWER TRACT) HAEMATURIA PATTERN

	Glomerular	Urological
Appearance	Dark, tea-coloured	Bright red ± clots; isolated terminal
Microscopy/ dipstick	Dysmorphic RBCs or casts + proteinuria	No proteinuria, infection or trauma
Examples	Post-streptococcal glomerulonephritis	Benign idiopathic urethrorrhagia ; urethral stricture (if abnormal stream)

	Glomerular	Urological
Next tests	C3, C4, antinuclear antibody (ANA), antistreptolysin O (ASO) titre	Imaging; persistent with normal imaging → cystoscopy + bilateral retrograde pyelograms

- **Clots after bone marrow transplant** → haemorrhagic cystitis from **BK virus, cytomegalovirus, adenovirus**; after **cyclophosphamide** → haemorrhagic cystitis needing **more emergent** management.
- **Flank trauma/deceleration** → emergent exclusion of renal injury; lower tract injury if pelvic fracture or genital laceration.
- As indicated: full blood count, electrolytes, creatinine, viral/bacterial testing.

Renal tumour

- Presentation: **palpable, firm, non-tender abdominal mass that rarely crosses the midline**.
- **Wilms tumour = commonest renal cancer <15 years; peak 1–5 years; ≈10% syndromic**.
- **Differential: congenital mesoblastic nephroma (neonate)**, neuroblastoma, renal cell carcinoma, clear cell sarcoma, rhabdoid tumour, **renal medullary carcinoma (sickle cell trait/disease)**, metastasis, leukaemia/lymphoma (management: Chapter 63).

Testicular tumour

- **≈1% of paediatric solid tumours** (0.5–2 per 100,000); **bimodal — <3 years and postpubertal**.
- Usually **painless mass**; keep suspicion in every scrotal examination, especially **testicular size asymmetry**; palpate each testis (consistency, size, position) and paratesticular structures.
- Mass found → look for **metastasis**: abdominal mass, leg oedema, supraclavicular nodes. **Precocious puberty** (pubic hair, penile enlargement) → **Leydig cell tumour**.

COMPARISON — PREPUBERTAL VS POSTPUBERTAL TESTICULAR MASSES

Prepubertal	Postpubertal
Usually benign — teratoma, Leydig cell tumour, epidermoid cyst; commonest malignancy yolk sac tumour	More likely malignant

- **Scrotal ultrasound with Doppler** confirms solid mass; also used when examination limited, **size discrepancy without palpable mass** (non-palpable intratesticular lesion) or **tense hydrocele**.
- **Markers promptly: α-fetoprotein (AFP), β-human chorionic gonadotrophin (β-hCG), lactate dehydrogenase (LDH)** → inform **testis-sparing surgery** discussion (with frozen-section support).
- Leydig cell suspicion/precocity → **testosterone, follicle-stimulating hormone (FSH), luteinising hormone (LH), prolactin** (management: Chapter 64).

Semiurgent referrals

Primary amenorrhoea

- **Definition**: no menses by **16 years with normal secondary sexual development**, or by **14 years with none**.
- History/examination: secondary sexual characteristics, family history, **cyclic abdominal pain**.
- **Differential**: constitutional delay; hypogonadotropic hypogonadism; chronic illness (e.g. hypothyroidism); **hypergonadotropic hypogonadism (primary ovarian failure, e.g. gonadal dysgenesis)**; pregnancy; androgen insensitivity syndrome (AIS); Müllerian agenesis; polycystic ovary syndrome.
- **Cyclic pain** → **genital outflow obstruction** (transverse vaginal septum, imperforate hymen).
- Initial evaluation: **are Müllerian structures present?** (examination, pelvic ultrasound, endoscopy) + **FSH, LH ± karyotype**.

Interlabial mass

- Mass appearing to **protrude from the introitus**.
- **Neonate:** periurethral/Skene gland, Bartholin or Gartner duct cysts; **ureterocele; imperforate hymen; rhabdomyosarcoma**. **Older child:** add **urethral prolapse**.
- **Most urgent step: exclude upstream obstruction** — pelvic/renal ultrasound for **hydrometrocolpos** (imperforate hymen) or **hydronephrosis** (prolapsed ureterocele); ultrasound also stages rhabdomyosarcoma (Chapter 58).

Routine referrals

Asymptomatic hernia or hydrocele

- **Indirect inguinal hernia** → inguinal bulge, inguinoscrotal swelling or hydrocele; **commoner after prematurity**.
- **Pathology: patent processus vaginalis** → fluid (hydrocele) or abdominal contents (hernia) between visceral and parietal tunica vaginalis.
- Swelling ↑ with crying, vomiting, constipation, cough. **Fluctuating size → communicating**; classically **smaller in the morning** after lying supine.
- Normal clinic examination → provoke with **Valsalva** (sit-up, blowing bubbles, standing with abdominal flexion). A **reliable parental history of intermittent swelling** suffices for diagnosis.
- **Operate if:** hernia on examination, **or compelling history with normal examination** (parental history, prior positive examination, photograph).
- Communicating hydrocele and hernia → same operation: **high ligation of processus vaginalis at the internal ring** (Chapter 55).
- **Hernia precautions** for any hernia, hydrocele or undescended testis: pain, **firm/non-reducible bulge**, erythema, inconsolable infant ± vomiting.
- **Resolved incarceration or bowel-containing hernia → timely surgery**.

Undescended testicle

- Detail: Chapter 57. **Empty hemiscrotum at birth** = undescended testis **or in utero torsion**.
- Descended at birth, **newly empty later** → **retractile vs ascending testis**.
- History: age first noticed, prematurity, inguinal bulge, prior groin surgery, fluctuating size; often **patent processus** → **hernia risk**.
- Inspect for **micropenis, hypospadias, chordee** → abnormal findings raise **DSD** suspicion.
- **Systematic palpation:** inguinal canal, mons, scrotum, perineum, upper thigh (**ectopic testis**).
- **Ultrasound not recommended** — poor sensitivity and specificity for location.

Varicocele

- **≈15% of adolescents**; parental anxiety about fertility; **mostly incidental**; few with pain/heaviness; may cause **testicular size asymmetry**.
- Examination: **testicular volume, consistency, Tanner stage, varicocele grade**, supine and standing; teach **self-examination**.
- Objective volume by **orchidometer or ultrasound**; **Lambert formula** = length × width × height × 0.71.
- **Non-decompressing supine**, especially **right-sided** → **retroperitoneal ultrasound** to exclude tumour compressing the gonadal vein.
- Management of asymptomatic adolescent varicocele is **controversial** (Chapter 55).

Phimosis

COMPARISON — PHYSIOLOGICAL VS PATHOLOGICAL PHIMOSIS

	Physiological	Pathological
Age	Neonates, early childhood; resolves with age ($\approx 1\%$ at 17 years)	Older children
Features	Non-retractile, asymptomatic; smegma	Erythema, swelling, pain, dysuria, ballooning ; constrictive band/ cicatrix
Management	Parental education; never forcibly retract	Treat; lichen sclerosus (balanitis xerotica obliterans, BXO) — thickening, hypopigmentation — must be treated

- **Treat asymptomatic physiological phimosis** in infants at **higher UTI risk** (VUR, prior UTI); circumcision declined → **topical steroids** reasonable.
- Urine trapping → **balanitis** (glans) or **balanoposthitis** (prepuce + glans) → seek treatment to prevent recurrence.
- **Smegma**: white/yellow subpreputial material, especially at coronal sulcus; **resolves when foreskin retracts**; seen in $\approx 5\%$ of healthy children.

Meatal stenosis

- Almost exclusively **circumcised males**.
- Symptoms: **dorsally deflected**, thin or forceful stream, spraying, split stream, dysuria, haematuria, hesitancy, frequency, distal penile pain; some asymptomatic.
- **Diagnosis on examination: meatus <5 Fr**; some have a **ventral web**.
- Uroflowmetry: **plateaued or peaked curve**; incomplete emptying, even retention. Long-standing → bladder wall thickening/trabeculation.
- **Symptomatic → meatotomy**.

Labial adhesions

- **Commonest interlabial abnormality; peak 13–23 months; rare in newborns** (maternal oestrogen).
- **Asymptomatic → observe**; spontaneous resolution with age.
- Symptoms: vaginal pooling, recurrent UTI, deflected stream, hesitancy, pain/itch, **post-void dribbling on standing** (vaginal voiding — evacuation of retained vaginal urine when rising).
- **Examine in frog-leg/butterfly position**; to see introitus, **pull labia majora straight out**, not laterally → vulva funnels open without tension.
- Treatment if symptomatic/desired: teach caregivers positioning → **topical oestrogen, steroid or petroleum-based product**; butterfly position after bathing to check recurrence without touching; **manual lysis** under topical anaesthetic or sedation.
- **Recurrence 7–55%**.

Figure 35.4 — Labial adhesions. Fusion of the **labia minora from the posterior fourchette to the clitoris**.

Bladder bowel dysfunction

- **Bladder bowel dysfunction (BBD)**: spectrum of impaired storage/elimination of **urine and stool**.
- Paediatric lower urinary tract symptoms have **different aetiologies from adults**: frequency, urgency, incontinence, enuresis, delayed toilet training, straining, recurrent non-febrile UTI, genital irritation/vaginitis, haematuria, **encopresis, constipation**.
- Treatment: behavioural → medical → biofeedback → pelvic floor physiotherapy → psychological care; **rarely surgery**.

- **History from child and guardian** (perspectives differ); **validated questionnaires** give an objective measure over time — Dysfunctional Voiding Scoring System, Dysfunctional Voiding and Incontinence Scoring System, Incontinence Symptom Index, Vancouver Scale, ICIQ-Pediatric LUTS, Childhood Bladder and Bowel Dysfunction Questionnaire, Iowa Voiding Questionnaire.
- Usually **after toilet training**; associated with **behavioural diagnoses**.
- **Exclude occult spinal dysraphism**: leg weakness/sensory change; **gluteal cleft asymmetry, sacral dimple, hair tuft, haemangioma** → spinal imaging or neurosurgery; gait abnormality/toe walking; sensory level if wheelchair-bound; scoliosis (Chapter 47).

Microscopic haematuria

- **Dipstick blood must be confirmed by microscopy**, and **UTI excluded**, before diagnosis.
- **Definition**: **>5 RBCs/high-power field (hpf)** on centrifuged urine on **more than two occasions**.
- Symptoms guide referral: urology, nephrology or both; wide differential (online eTable 35.2).
- Seek family renal history, immunological history, **hypertension**, crystals, **proteinuria**, **casts**.
- **Mostly benign** → **observe and retest**; screening renal-bladder ultrasound common but its **utility is debated**.

KEY POINTS

- **Spermatic cord torsion** is a true emergency and a clinical diagnosis; if strongly suspected, defer imaging and explore immediately.
- **Ambiguous genitalia in the newborn** is an emergency — exclude life-threatening conditions, establish the diagnosis and sex of rearing; a multidisciplinary team is standard of care.
- **Gross haematuria** differential is narrowed by character (tea-coloured vs bright red) and associated symptoms (dysuria, flank pain, recent illness).
- **BBD** is categorised and treatment focused using voiding questionnaires.

Concept — *triage asks one question: **what is lost by waiting?** Torsion, salt-wasting CAH, infected obstruction and paraphimosis cost an organ or a life within hours; children also **hide shock until it decompensates**, so clinical suspicion, not imaging or blood pressure, drives the emergent tier.*

EVALUATION OF THE PAEDIATRIC PATIENT

History

- **Engage the child, set expectations**; dispel fears of surgery/examination so the child participates.
- **Privacy for older children**: sexual/social history may be taken **alone**; ask preferences at the start; explain prior traumatic tests and what to expect.
- **Identify everyone present** and their relationship (social worker, foster parent); others may leave.
- **Prenatal history** (most relevant to hydronephrosis, not testicular complaints): natural conception vs **in vitro fertilisation (IVF)**; intrauterine growth restriction, maternal diabetes; prenatal anomalies — cardiac, spinal dysraphism, ambiguous genitalia, **discordant cell-free DNA/karyotype/ultrasound genital appearance**, imperforate anus, omphalocele, abdominal cyst, Müllerian/cloacal anomalies.
- **Serial prenatal ultrasound** — gestation at detection, change with growth, **amniotic fluid volume** → proxy for fetal renal function (amniotic fluid is largely **fetal urine**).
- Surgical history may start **prenatally** (fetal dysraphism repair, **vesicoamniotic shunt**); some families meet urology antenatally.

- Prenatal exposures: maternal drugs, environment, illness, teratogens, **folic acid deficiency**; amniocentesis, chorionic villus sampling, genetic tests.
- **Birth history for all**: gestational age; neonatal intensive care unit (NICU) stay (anomalies, feeding, respiratory support); procedures (circumcision).
- Medications, family history and **siblings' surgical history** — some conditions are familial.
- **Social history**: medical power of attorney, foster care, caseworker, care facility; daycare/school; household; **who holds legal consent**.
- **Consent from guardian, assent from child** — age- and procedure-appropriate agreement matters.
- **All carers** need education and supplies (catheterisation, postoperative care), sometimes across homes; **sports seasons** affect timing/restrictions.
- **Safety**: bullying/abuse may surface during sexual or voiding history; understand **gender identity** as the child develops.
- **Developmental delay**; in neurogenic bladder — **mobility, hand dexterity, sensory level** (self-catheterisation, independent transfer).
- **Comorbid psychological conditions** — attention-deficit/hyperactivity disorder (ADHD), anxiety, depression, autism, intellectual disability — matter in bladder/bowel and chronic complaints.

Physical examination

Paediatric vital signs

- Norms **vary with age**; distress and wrong-sized equipment distort readings.

TABLE 35.3 NORMAL PEDIATRIC VITAL SIGNS ACCORDING TO AGE

Age	Heart rate (beats/min)	Blood pressure (mm Hg, systolic/diastolic)	Respiratory rate (breaths/min)
Premature	120–170	55–75/35–45	40–70
0–3 mo	100–150	65–85/45–55	35–55
3–6 mo	90–120	70–90/50–65	30–45
6–12 mo	80–120	80–100/55–65	25–40
1–3 yr	70–110	90–105/55–70	20–30
3–6 yr	65–110	95–110/60–75	20–25
6–12 yr	60–95	100–120/60–75	14–22
12+ yr	55–85	110–135/65–85	12–18

Modified from Hartman and Cheifetz, Nelson Textbook of Pediatrics, 2009.

- **Pattern**: heart and respiratory rates **fall**, blood pressure **rises** with age.
- **Rules of thumb**: respiratory rate should not stay **>60/min**; **heart rate** $\approx 2\text{--}3 \times$ **respiratory rate**.
- **Lower limit of systolic BP**: ≥ 60 neonate; ≥ 70 1 month–1 year; $\geq 70 + (2 \times \text{age})$ 1–10 years; ≥ 90 >10 years.
- **Cuff should cover** $\approx$ **two-thirds of the arm**: **too small** $\rightarrow$ **falsely high**, **too large** $\rightarrow$ **falsely low**.
- **Trap**: shock presents first as **tachypnoea and tachycardia**; **hypotension is not required** and usually means **advanced, decompensated shock**.
- **Weight every visit** (drug dosing); growth charts and body mass index (BMI) detect failure to thrive, malnutrition, obesity — key in CKD, stones, spina bifida, endocrine disorders.
- **BMI** = **weight (kg)/height (m)²**; percentiles for ≥ 2 years, weight-for-length for <2 years. **Underweight** $<5\text{th}$; **overweight** 85th–95th; **obese** $\geq 95\text{th}$ percentile.

Abdominal examination

- Distinguishes abdominal pain causes; detects **distended bladder**, **constipation**, stigmata of chronic disease (**lax abdominal wall — prune belly syndrome**).
- **Costovertebral angle tenderness** → pyelonephritis or renal colic. In infants, **renal mass or severe hydronephrosis** often palpable.

Male genital examination

- **Explain first**; tell young children only parents and doctors (parent present) may examine.
- Adolescents: **privacy, gowns**. Prepubertal: **parent at the head**. Parent absent → **documented chaperone**.
- **Penis**: foreskin retractability (uncircumcised); **adhesions and skin bridges** (circumcised); other anomalies.
- **Scrotum/inguinal canal**: testicular position, consistency, size, symmetry; hydrocele, mass, bulge. **Prepubertal** → **supine/frog-leg**; adolescent → **standing**; **Valsalva** for hernia or varicocele.
- **Tanner staging** (five stages): males by **genitalia + pubic hair**, females by **breast + pubic hair** → guides delayed/precocious puberty differentials.

TABLE 35.4 TANNER STAGING FOR MALES AND FEMALES

Stage	Male — genitalia and pubic hair	Female — pubic hair
I (preadolescent)	Testes, scrotum, penis as early childhood; no pubic hair	No sexual hair
II	Scrotal and testicular enlargement , thinning scrotal skin; scant hair	Sparse, downy hair mainly along labia majora
III	Penile growth — length first, then girth ; testes/scrotum grow; hair may curl	Darker, coarser, curlier hair spread sparsely over pubis
IV	Penis longer and broader, glans develops; scrotal skin darkens; coarse curly hair, less than adult	Adult-type but less quantity; no spread to thighs
V	Adult size/shape; hair spreads to medial thighs	Adult inverse triangle ; spreads to medial thighs

Data modified from Escobar et al., Zitelli and Davis' Atlas of Pediatric Physical Diagnosis, 2023; Palmer and Palmer, Campbell-Walsh-Wein Urology, 2021.

- **Pattern**: in boys the **first sign of puberty is testicular/scrotal enlargement (II)**, not hair; **spread of hair to the thighs defines stage V** in both sexes.
- Table gives the female pubic-hair scale only; female breast staging is referred to in the text but not tabulated.
- **Micropenis** is still diagnosed against **1940s norms** (Table 35.5), which ignore **ethnicity and gestational age** — the main determinants of length.
- **Stretched penile length**: from **pubic symphysis attachment to glans tip**, on stretch.

TABLE 35.5 STRETCHED PENILE LENGTH (CM) IN NORMAL MALE SUBJECTS

Age	Mean ± SD	Mean – 2.5 SD
Neonate, 30-wk gestation	2.5 ± 0.4	1.5
Neonate, 34-wk gestation	3.0 ± 0.4	2.0
0–5 mo	3.9 ± 0.8	1.9
6–12 mo	4.3 ± 0.8	2.3
1–2 yr	4.7 ± 0.8	2.6
2–3 yr	5.1 ± 0.9	2.9

Age	Mean $\pm$ SD	Mean - 2.5 SD
3-4 yr	5.5 $\pm$ 0.9	3.3
4-5 yr	5.7 $\pm$ 0.9	3.5
5-6 yr	6.0 $\pm$ 0.9	3.8
6-7 yr	6.1 $\pm$ 0.9	3.9
7-8 yr	6.2 $\pm$ 1.0	3.7
8-9 yr	6.3 $\pm$ 1.0	3.8
9-10 yr	6.3 $\pm$ 1.0	3.8
10-11 yr	6.4 $\pm$ 1.1	3.7
Adult	13.3 $\pm$ 1.6	9.3

Data from Feldman and Smith, 1975; Schonfeld and Beebe, 1942; Tuladhar et al., 1998.

- **What it is for:** micropenis = stretched length $<$ mean - 2.5 SD for age — term newborn $\approx$ 1.9 cm, adult $\approx$ 9.3 cm.
- **Pattern:** length barely changes from $\approx$ 5 to 11 years ($\approx$ 6 cm mean) — growth is pubertal; preterm infants need **gestation-specific** values.
- **Undescended testis examination** (provider-specific; see Video 35.1): **frog-leg, relaxed** $\rightarrow$ sweep one hand from **anterior superior iliac spine along the inguinal canal** $\rightarrow$ other hand grasps testis and brings it to dependent scrotum; **soap/water or lubricant** reduces friction.
- Older or obese child: testis may become palpable **sitting cross-legged or in the squatting “catcher's” position**.

Female genital examination

- **Frog-leg position;** adolescents sometimes **stirrups** with draping.
- Visualise with **caudal traction on labia majora away from the body**, minimal lateral traction.
- Inspect labia majora/minora, clitoris, urethral meatus, introitus, anus; **Tanner stage, labial adhesions, discharge, urine pooling** (Video 35.2).

Neurologic examination

- **Observe** mobility and interaction. Neonates/infants: **limb tone, head circumference**. $\geq$ 12 months: play, **gait**, movement symmetry.
- Spinal dysraphism (**myelomeningocele, tethered cord**) $\leftrightarrow$ neurogenic bladder and bowel $\rightarrow$ inspect **lumbosacral skin** for occult dysraphism.
- High- vs low-risk markers are **poorly defined**; predictive data lacking.
- “Simple dimple” = **within the gluteal cleft, $<$ 5 mm, base clearly seen** $\rightarrow$ **low correlation with pathology**.
- **Neurosurgeon consensus** (asymptomatic child): **palpable lipoma** $\rightarrow$ **image** (unanimous); **simple dimple** $\rightarrow$ **observe without imaging** unless symptomatic; **image** for coccygeal tag $>$ 5 mm, deep dimple, atypical dimple, deviated gluteal crease. **Sacral dimples are over-investigated and over-referred**; poor agreement on gluteal cleft deviation and cutaneous vascular marks.

Figure 35.5 — Lumbosacral skin stigmata in infants (Ben-Sira et al., 2009).

- (A) **Simple dimple $<$ 25 mm above the anus — low risk.** (B) **Symmetrical deviated gluteal fold — low risk.** (C) **Gluteal fold deviated by an underlying mass — not low risk.**

Laboratory studies

Urine studies

Urinary tract infection.

- Collection is hard in the **non-toilet-trained** child; dipstick, microscopy and culture all contribute.
- **American Academy of Pediatrics (AAP) 2011 guideline** (under revision): **urinalysis with pyuria and/or bacteriuria + $\geq 50,000$ colony-forming units/mL of a uropathogen** from a **catheterised or suprapubic aspirate** specimen.
- **Contamination rises as invasiveness falls**; AAP → **catheter or suprapubic aspirate for both urinalysis and culture in febrile infants <24 months** — **bag specimens give very high false positives** (methods: Chapter 37).
- Collection route **matters less when not testing for infection** (haematuria, proteinuria).

Haematuria.

- Microhaematuria on screening in **0.5–3%** of children; **gross haematuria rarer → more aggressive evaluation**.
- **Isolated microhaematuria usually self-limiting and benign**; three commonest causes: **thin basement membrane disease, immunoglobulin A (IgA) nephropathy, hypercalciuria**.
- **Dipstick blood → always confirm by microscopy** — false positives from **haemoglobinuria and myoglobinuria**.
- Microhaematuria → **repeat over 2–3 weeks** → persistent → **imaging**.
- **Gross or persistent microhaematuria → random urine calcium/creatinine**. Hypercalciuria → **high fluid intake + modest sodium restriction**; it is a stone risk factor → **renal ultrasound and urology referral**.

Proteinuria.

- With haematuria → suggests **glomerular cause**. Many RBCs can falsely raise dipstick protein (lysis releases haemoglobin).
- Reflects **disrupted glomerular filtration barrier**: **transient** (fever, strenuous exercise), **persistent benign (orthostatic)**, or **persistent from renal disease**.
- **First-morning samples** because orthostatic proteinuria is common. Low-level dipstick protein in healthy children is usually transient/false positive; **AAP does not recommend screening**.
- Confirm with **random urine protein/creatinine**. **Persistent non-orthostatic proteinuria on two early-morning samples = pathological until proven otherwise → nephrology**.
- Paediatric differential skews to **congenital renal dysplasia (reflux nephropathy) and obstructive uropathy**, as well as glomerular disease.

Specific gravity. Reflects concentrating ability and hydration; physiological range **1.002–≈1.040**.

COMPARISON — PAEDIATRIC URINE THRESHOLDS

Test	Threshold	Meaning
Microscopic haematuria	>5 RBCs/hpf (child) vs ≥ 3 RBCs/hpf (adult), without infection	No accepted paediatric guideline; child threshold is higher
Calcium/creatinine (random)	>0.4 age 2–5 yr; >0.2 age >5 yr	Hypercalciuria
Protein/creatinine — normal	<0.7 first year; <0.4 up to 4 years; <0.2 after 5 years	Normal falls with age

Test	Threshold	Meaning
Protein/creatinine — nephrotic	>3.0	Nephrotic range
Specific gravity	<1.010 / 1.010–1.020 / >1.020	Excess water intake or poor concentration (diabetes insipidus) / typical normal random sample / dehydration, low intravascular volume
Urine culture (AAP)	≥50,000 CFU/mL, catheter or suprapubic aspirate	UTI with abnormal urinalysis

- **Pattern:** normal ratios are **higher the younger the child**; adult cut-offs mislabel infants.

Serum studies

Complete blood count.

- **Anaemia prevalent in CKD**; haemoglobin falls once glomerular filtration rate (GFR) <43 mL/min/1.73 m².
- Paediatric reference values matter for **postoperative bleeding**. **Post-newborn clamp circumcision bleeding** is rare but can need admission, transfusion, reoperation.
- **Trap:** haemoglobin may be **normal initially after acute bleeding, then fall precipitously**. Neonates tolerate hypovolaemia **worse than adults** → watch **pallor, shallow/rapid breathing, weak or absent peripheral pulses**.

Serum creatinine and cystatin C.

- Both predict GFR, **alone or combined**; best paediatric method still under study, especially with comorbidity.
- **Ideal GFR marker: filtered, not secreted or reabsorbed** by tubules.

Urologic imaging

- Overview of modalities in Chapter 36. **ALARA** (“as low as reasonably achievable”) is paramount; paediatric dose is **hard to quantify** (variable technique, size, age).

TABLE 35.7 COMPARISON OF PEDIATRIC UROLOGY IMAGING MODALITIES

Modality	Uses in paediatric urology	Advantages	Disadvantages
Ultrasound	Renal/bladder; scrotal/testicular; testicular torsion ; prenatal; spinal	Inexpensive; no sedation; real-time; no contrast	Equipment- and operator-dependent
Radiography/fluoroscopy*	Bony abnormality; radiopaque stones; stool burden; genitogram; urodynamics; VCUG (VUR)	Wide range of indications; real-time	Ionising radiation
Computed tomography*	Severe blunt abdominal trauma ; nephrolithiasis; renal artery disease	Widely available; fast; no sedation	High radiation dose
Magnetic resonance imaging	Renal/bladder tumours; pelvicaliectasis; genitography; urogenital sinus; ureteral ectopia ; duplication; Müllerian anomalies	Excellent anatomy ; urogram gives function	Sedation in young children ; limited availability; high cost
MAG-3 renography* (mercaptoacetyl triglycine)	Obstruction — UPJ, UVJ	Obstruction + differential function	Ionising radiation
DMSA scintigraphy* (dimercaptosuccinic acid)	Pyelonephritis/parenchymal damage ; function of moieties in duplex systems	Scarring + differential function	Ionising radiation

*Ionising radiation. Modified from Coco and Peters, Campbell-Walsh-Wein Handbook of Urology, 2022.

- **What it is for:** matching the question to the modality — **anatomy (ultrasound, MRI), dynamic voiding (VCUG), drainage (MAG-3), cortical scar (DMSA).**
- **Pattern:** only **ultrasound and MRI avoid ionising radiation**; MRI trades radiation for **sedation**, CT trades sedation for **high dose**.
- **Trap:** both nuclear scans give differential function, but **MAG-3 answers obstruction** and **DMSA answers scarring**.

KEY POINTS

- **Paediatric history** is directed at both child and caregiver; set expectations and goals up front.
- **Frog-leg or butterfly position** favours genital examination in prepubertal males and females.
- **Simple sacral dimples** generally need no magnetic resonance imaging (MRI) or neurosurgical consultation.
- **AAP diagnosis of UTI** requires urinalysis evidence of infection (pyuria and/or bacteriuria) with a positive culture in a symptomatic child.
- **Minimise radiation (ALARA)** whenever imaging a child.

Concept — every paediatric norm is **age-indexed**: vital signs, bladder capacity, penile length, urine ratios and BMI all move with age, so an adult threshold applied to a child is the commonest error — and the child's **physiological reserve masks deterioration** until late.

MULTIDISCIPLINARY CARE

- Needs arise from the **psychological impact of diagnosis** and **complex underlying pathology**.
- **Goals:** centralise care, improve efficiency and access, foster innovation for complex pathology.
- **Forms:** **multidisciplinary clinic** (several providers in one visit) or **multidisciplinary conference** (plan relayed to family).
- **Established areas:** prenatal diagnosis, spina bifida, nephrolithiasis, BBD, DSD, complex pelvic anomalies; radiology and pathology input adds value.

SPECIAL CONSIDERATIONS FOR CHILD COMFORT WITH EVALUATION

- Children often **cannot understand the reason** for intervention and react to separation, pain, anxiety → adult awake procedures (**catheterisation, flexible cystoscopy, priapism treatment**) often not feasible awake.
- **Distraction and child life specialists:** tablet ≈ midazolam for presurgical anxiety, emergence delirium, length of stay; **virtual reality** ↓ preoperative anxiety; extendable to catheterisation, VCUG, urodynamics, office procedures.
- **Preprocedural sedation** (anxiolytic, sedative, analgesic) can **avoid general anaesthesia** for minor procedures.

COMPARISON — AGENTS FOR PAEDIATRIC PROCEDURAL COMFORT

Agent	Route	Use	Evidence/bottom line
Midazolam	Premedication	Before VCUG ; presurgical anxiety	↓ recall and anxiety without compromising VCUG results ; equivalent to tablet distraction

Agent	Route	Use	Evidence/bottom line
Ketamine	Intranasal	Procedural sedation/analgesia	Systematic review: safe and effective ; superior to comparators (e.g. midazolam) in some, not all, studies
LMX (lidocaine 4%)	Topical	Office procedures	Adequate analgesia for in-office meatotomy
EMLA (lidocaine 2.5% + prilocaine 2.5%)	Topical, 30–45 min before	Lysis of penile adhesions/skin bridges, labial adhesions, meatotomy, excision of benign lesions (suture sinus)	Trial: EMLA + midazolam or nitrous oxide ≈ general anaesthesia for meatotomy (pain, complications, success); large review: well tolerated, effective, safe, avoids GA
Lidocaine gel	Urethral	Awake urethral catheterisation	Mixed results

- **Pattern:** topical anaesthesia works well for **skin-level office procedures**; the evidence weakens once the procedure is **intraurethral** (lidocaine gel).

Concept — in children the **limiting factor is cooperation, not technique**: distraction, sedation and topical anaesthesia are escalating substitutes for the understanding a child lacks, each chosen to **avoid a general anaesthetic** the procedure itself does not require.